- One particle (Conservative Force)



find $\frac{d}{dr}\int_{t_1}^{t_2} \underbrace{\frac{1}{2}m\dot{r}(t)^2 - U(r(t))}\, dt = 0$

Lagrangian = kinetic energy − potential energy

$L(r,\dot{r},t) = \frac{1}{2}m\dot{r}(t) - U(r(t))$

From Euler–Lagrange in Calculus of Variation $\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} - \frac{\partial L}{\partial r} = \underbrace{m\ddot{r}(t) + \nabla U(r(t))}_{F=ma} = 0$

Coordinate-Invariance

find q $\frac{d}{dt}\int_{t_1}^{t_2} \frac{1}{2}m\|Dr(q)\dot{q}\|^2 - U(r(q))\, dt = 0$

$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = 0$ Euler–Lagrange still holds if we use generalized coordinate

$$\begin{aligned}
&\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} \\
&= \frac{d}{dt}\left[m\,Dr(q)^T Dr(q)\dot{q}\right] - m\frac{\partial}{\partial q}\left(Dr(q)\dot{q}\right)^T Dr(q)\dot{q} + Dr(q)^T\nabla U(r(q)) \\
&= m\frac{d}{dt}\left[\nabla r_1(q) \cdots \nabla r_n(q)\right] Dr(q)\dot{q} + m\,Dr(q)^T\frac{d}{dt}\left(Dr(q)\dot{q}\right) - m\left(\frac{\partial}{\partial q}\begin{bmatrix} -\nabla r_1(q)^T\dot{q}- \\ \vdots \\ -\nabla r_n(q)^T\dot{q}- \end{bmatrix}\right)^T Dr(q)\dot{q} + Dr(q)^T\nabla_r U(r(q)) \\
&= m\left[\nabla r_1\dot{q} \cdots \nabla r_n\dot{q}\right] Dr(q)\dot{q} + m\,Dr(q)^T\ddot{r} - m\begin{bmatrix} -\nabla r_1(q)^T\dot{q}- \\ \vdots \\ -\nabla r_n(q)^T\dot{q}- \end{bmatrix} Dr(q)\dot{q} + Dr(q)^T\nabla_r U(r(q)) \\
&= Dr(q)^T\left(m\ddot{r} + \nabla_r U(r(q))\right)
\end{aligned}$$

- Multiple Particle (Conservative Force)

find q $\frac{d}{dq}\int_{t_1}^{t_2} \sum_{i=1}^{N} \frac{1}{2}m_i\|Dr_i(q)\dot{q}\|^2 - U(r_1(q)\cdots r_N(q))\, dt = 0$

$$\begin{aligned}
&\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} \\
&= \frac{d}{dt}\left[\sum_{i=1}^{N} m_i\, Dr_i(q)^T Dr_i(q)\dot{q}\right] - \sum_{i=1}^{N} m_i\frac{\partial}{\partial q}\left(Dr_i(q)\dot{q}\right)^T Dr_i(q)\dot{q} + \sum_{i=1}^{N} Dr_i(q)^T\nabla_{r_i}U \\
&= \sum_{i=1}^{N} m_i\frac{d}{dt}\left[\nabla r_i^1(q) \cdots \nabla r_i^n(q)\right] Dr_i(q)\dot{q} + m_i\,Dr_i(q)^T\frac{d}{dt}\left(Dr_i(q)\dot{q}\right) - \sum_{i=1}^{N} m_i\left(\frac{\partial}{\partial q}\begin{bmatrix} -\nabla r_i(q)^T\dot{q}- \\ \vdots \\ -\nabla r_{in}(q)^T\dot{q}- \end{bmatrix}\right)^T Dr_i(q)\dot{q} + \sum_{i=1}^{N} Dr_i(q)^T\nabla_{r_i}U \\
&= \sum_{i=1}^{N} m_i\, Dr_i(q)^T\ddot{r}_i + \sum_{i=1}^{N} Dr_i(q)^T\nabla_{r_i}U \\
&= \sum_{i=1}^{N} Dr_i(q)^T\left[m_i\ddot{r}_i + \underbrace{\nabla_{r_i}U}_{\text{active force}}\right] \\
&= 0 \text{ by D'Alembert principle}
\end{aligned}$$

- Principle of Least / Stationary Action

  action $= \int_{t_1}^{t_2} T(q,\dot{q}) - U(q)\, dt$

  along the trajectory, the system is losing potential energy and gaining kinetic energy (or the other way around)

  the trading is dominated by $\frac{d}{dt}(\frac{1}{2}mv^2) = ma^T v = F^T v = -\nabla_r U(r)^T v$

  when the trajectory is perturbed by an admissible perturbation δq

  the total change in potential energy is $\frac{d}{d\varepsilon}\int U(q)\, dt$, which is traded for kinetic energy $\frac{d}{d\varepsilon}\int T(q,\dot{q})\, dt$

$$\begin{aligned}
&\frac{d}{d\varepsilon}\int_{t_1}^{t_2} T(q,\dot{q}) - U(q)\, dt \\
&= \int_{t_1}^{t_2} \frac{\partial T}{\partial q}\delta q + \frac{\partial T}{\partial \dot{q}}\delta\dot{q} - \frac{\partial U}{\partial q}\delta q\, dt \\
&= \int_{t_1}^{t_2} \frac{\partial L}{\partial q}\delta q + \frac{\partial T}{\partial \dot{q}}\delta q\Big|_{t=t_1}^{t=t_2} - \int \delta q\, d\frac{\partial T}{\partial \dot{q}} \\
&= \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)\delta q\, dt \\
&= 0
\end{aligned}$$

$\delta q(t_1) = 0$ Starting point is known

$\delta q(t_2) = 0$ assuming we know end point

- Non-Conservative Force

  $\frac{d}{d\varepsilon}\int T(q,\dot{q}) - U(q)\, dt = \int \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)\delta q = -\int \sum_{i=1}^{N} F_{i,nc}^{(ace)\,T} Dr_i(q)\, \delta q\, dt$

  non-conservative force can compensate discrepancy between $\frac{d}{d\varepsilon}\int T(q,\dot{q})\, dt$ and $\frac{d}{d\varepsilon}\int U(q)\, dt$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(ace)}$$

- Holonomic Constraints

  $\text{find}_{\delta q}$ $\frac{d}{d\varepsilon}\int T(q,\dot{q}) - U(q)\, dt = -\int \sum_{i=1}^{N} F_{i,nc}^{(ace)\,T} Dr_i(q)\, \delta q\, dt$ $\quad \forall$ admissible δq

  s.t. $h(q) = 0 \ \forall t$ work done by non-conservative force through admissible displacement δq

$$\begin{cases} \frac{d}{d\varepsilon}\int T(q,\dot{q}) - U(q)\, dt = \int \left(\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}}\right)\delta q\, dt = -\int \sum_{i=1}^{N} F_{i,nc}^{(ace)\,T} Dr_i(q)\, \delta q\, dt & \forall \text{ admissible } \delta q \\ \frac{\partial h}{\partial q}\delta q = 0 & \delta q \text{ is admissible} \end{cases}$$

$$\Downarrow$$

$$\begin{cases} \left(\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} - \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(ace)}\right)\delta q = 0 & \forall \text{ admissible } \delta q \\ \frac{\partial h}{\partial q}\delta q = 0 & \delta q \text{ is admissible} \end{cases}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \underbrace{\sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(ace)} + \overbrace{\frac{\partial h}{\partial q}^T}^{\text{constraint force}}\lambda}_{\text{total force}}$$

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(ace)} + \frac{\partial h}{\partial q}^T \lambda$$

need to solve for both $\ddot{q}(t)$ and $\lambda(t)$

$$\begin{aligned}\frac{d^2}{dt^2}h(q) &= \frac{d}{dt}\left(\frac{\partial h}{\partial q}\dot{q}\right)\\ &= \frac{d}{dt}\left(\begin{bmatrix}-\nabla h_1-\\ \vdots\\ -\nabla h_m-\end{bmatrix}\dot{q}\right)\\ &= \frac{\partial h}{\partial q}\ddot{q} + \begin{bmatrix}-\nabla^2 h_1(q)\dot{q}-\\ \vdots\\ -\nabla^2 h_m(q)\dot{q}-\end{bmatrix}\dot{q}\end{aligned}$$

$$\begin{cases}\dfrac{\partial h}{\partial q}\ddot{q} + \begin{bmatrix}-\nabla^2 h_1(q)\dot{q}-\\ \vdots\\ -\nabla^2 h_m(q)\dot{q}-\end{bmatrix}\dot{q} = 0\\ M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \sum_{i=1}^{N} Dr_i(q)^T F_{i,mc}^{(act)} + \dfrac{\partial h}{\partial q}^T\lambda\end{cases}$$

$$\begin{bmatrix}M(q) & -\frac{\partial h}{\partial q}^T\\ -\frac{\partial h}{\partial q} & 0\end{bmatrix}\begin{bmatrix}\ddot{q}\\ \lambda\end{bmatrix} = \begin{bmatrix}-C(q,\dot{q}) - G(q)\\ \begin{bmatrix}-\nabla^2 h_1(q)\dot{q}-\\ \vdots\\ -\nabla^2 h_m(q)\dot{q}-\end{bmatrix}\dot{q}\end{bmatrix}$$

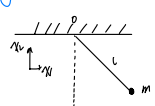
$$\frac{d^2}{dt^2}h(q) = \frac{\partial h}{\partial q}\ddot{q} + \cdots = \frac{\partial h}{\partial q}M(q)^{-1}\frac{\partial h}{\partial q}^T\lambda - \cdots$$

Baumgarte Stabilization:

first solve λ with block elimination

then augment λ with $\lambda := \lambda + (-\alpha h(q) - \beta \dot{h}(q))$ as PD controller

Eg. pendulum



$$q = \begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

$$L = \frac{1}{2}m(\dot{x}_1^2+\dot{x}_2^2) - mgx_2$$

$$h(q) = x_1^2 + x_2^2 - l^2 = 0$$

$$\left.\begin{aligned}&\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \begin{bmatrix}m & \\ & m\end{bmatrix}\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix} + \begin{bmatrix}0\\ mg\end{bmatrix}\\ &\sum_{i=1}^{N} F_{i,mc}^{(act)} + \frac{\partial h}{\partial q}^T\lambda = \begin{bmatrix}x_1\\ x_2\end{bmatrix}\lambda\end{aligned}\right\} \quad \underbrace{\begin{bmatrix}m & \\ & m\end{bmatrix}\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix} + \begin{bmatrix}0\\ mg\end{bmatrix}}_{M(q)\ddot{q} + a(q)} = \begin{bmatrix}x_1\\ x_2\end{bmatrix}\lambda$$

$$\left.\begin{aligned}&\frac{d}{dt}h(q) = \begin{bmatrix}x_1\\ x_2\end{bmatrix}^T\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix}\\ &\frac{d^2}{dt^2}h(q) = \begin{bmatrix}x_1\\ x_2\end{bmatrix}^T\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix} + 2(\dot{x}_1^2+\dot{x}_2^2)\end{aligned}\right\} \quad \begin{bmatrix}2x_1\\ 2x_2\end{bmatrix}\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix} = -2(\dot{x}_1^2+\dot{x}_2^2)$$

$$\begin{bmatrix}\begin{bmatrix}m & \\ & m\end{bmatrix}\begin{bmatrix}-x_1\\ -x_2\end{bmatrix}\\ \begin{bmatrix}-x_1 & -x_2\end{bmatrix} \quad 0\end{bmatrix}\begin{bmatrix}\begin{bmatrix}\ddot{x}_1\\ \ddot{x}_2\end{bmatrix}\\ \lambda\end{bmatrix} = \begin{bmatrix}\begin{bmatrix}0\\ -mg\end{bmatrix}\\ 2(\dot{x}_1^2+\dot{x}_2^2)\end{bmatrix}$$

- Pfaffian Constraints

write all constraints (holonomic or non-holonomic) in "pfaffian form"

$$A(q,t)^T \dot{q} + b(q,t) = 0$$

for holonomic constraints $\phi(q,t) = 0$

$$\frac{d}{dt}\phi(q,t) = \nabla_q \phi(q,t)^T \dot{q} + \frac{\partial}{\partial t}\phi(q,t) = 0$$

$$\text{find}_{q} \quad \frac{d}{dq}\int L(q,\dot{q})\,dt = \int \sum_{i=1}^{N} F_{i,nc}^{(act)^T} Dr_i(q)\,\delta q\,dt \quad \forall \text{ admissible } \delta q$$

$$\text{s.t.} \quad A(q,t)\dot{q} + b(q,t) = 0$$

$$\begin{cases} \int \left(\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} - \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(act)} \right) \delta q\,dt = 0 \quad \forall \text{ admissible } \delta q \\ A(q,t)\,\delta q = 0 \qquad \delta q \text{ is admissible} \end{cases}$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(act)} + A(q,t)^T \lambda$$

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(act)} + A(q,t)^T \lambda$$

need to solve for both $q(t)$ and $\lambda(t)$

$$\begin{cases} M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(act)} + A(q,t)^T \lambda \\ \frac{d}{dt}\left(A(q,t)\dot{q} + b(q,t) \right) = A(q,t)\ddot{q} + \frac{d}{dt}A(q,t)\,\dot{q} + \frac{d}{dt}b(q,t) = 0 \end{cases}$$

$$\begin{bmatrix} M(q) & -A(q,t)^T \\ -A(q,t) & 0 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ \lambda \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{N} Dr_i(q)^T F_{i,nc}^{(act)} - C(q,\dot{q}) - G(q) \\ -\frac{d}{dt}A(q,t)\,\dot{q} - \frac{d}{dt}b(q,t) \end{bmatrix}$$